

Отчет по лабораторной работе №2

Настройка DNS-сервера

Виноградова М.А

12 сентября 2025

Российский университет дружбы народов, Москва, Россия

Информация

- Виноградова Мария Андреевна
- студентка НПИбд-02-24
- номер студ. билета 1132240691
- Российский университет дружбы народов
- 1132240691@pfur.ru
- https://github.com/mavinogradova/study_2025-2026_os2

Цель работы

Приобретение практических навыков по установке и конфигурированию DNS-сервера, усвоение принципов работы системы доменных имён.

Задание

1. Установить на виртуальной машине server DNS-сервер bind и bind-utils.
2. Сконфигурировать на виртуальной машине server кэширующий DNS-сервер.
3. Сконфигурировать на виртуальной машине server первичный DNS-сервер.
4. При помощи утилит dig и host проанализировать работу DNS-сервера.
5. Написать скрипт для Vagrant, фиксирующий действия по установке и конфигурированию DNS-сервера во внутреннем окружении виртуальной машины server. Внести соответствующие изменения в Vagrantfile.

Выполнение лабораторной работы

Подключение к виртуальной машине server

Запускаем виртуальную машину server и подключаемся к ней по SSH.

```
D:\work\mavinogradova\vagrant>vagrant ssh server
==> server: The machine you're attempting to SSH into is configured to use
==> server: password-based authentication. Vagrant can't script entering the
==> server: password for you. If you're prompted for a password, please enter
==> server: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
Last login: Sat Sep 12 10:07:10 2026 from 10.0.2.2
vagrant@server:~$ sudo -i
```

Рис. 1: Подключение к server

Установка DNS-сервера

Входим под пользователем mavinogradova и переходим в режим суперпользователя.

```
vagrant@server:~$ su - mavinogradova
Password:
Last login: Sat Sep 12 10:08:07 UTC 2026 on pts/0
[mavinogradova@server.mavinogradova.net ~]$ dnf-y install bind bind-utils
bash: dnf-y: command not found...
[mavinogradova@server.mavinogradova.net ~]$ sudo -i
[sudo] password for mavinogradova:
```

Рис. 2: Вход под mavinogradova и получение root

Устанавливаем bind и bind-utils.

```
[root@server.mavinoegradova.net ~]# dnf -y install bind bind-utils
Last metadata expiration check: 0:06:04 ago on Sat 12 Sep 2026 10:35:43 AM UTC.
Package bind-utils-32:9.18.33-15.el10_2.10.x86_64 is already installed.
Dependencies resolved.
=====
Package                Architecture      Version              Repository           Size
=====
Installing:
bind                   x86_64            32:9.18.33-15.el10_2.10  appstream            323 k
Installing weak dependencies:
bind-dnssec-utils      x86_64            32:9.18.33-15.el10_2.10  appstream            147 k
Transaction Summary
=====
Install 2 Packages

Total download size: 470 k
Installed size: 1.3 M
Downloading Packages:
(1/2): bind-9.18.33-15.el10_2.10.x86_64.rpm                809 kB/s | 323 kB  00:00
(2/2): bind-dnssec-utils-9.18.33-15.el10_2.10.x86_64.rpm    241 kB/s | 147 kB  00:00
-----
Total                                                         76 kB/s | 470 kB  00:06
Running transaction check
```

Рис. 3: Установка bind и bind-utils

Выполняем запрос dig www.yandex.ru.

```
[root@server.mavinogradova.net ~]# dig www.yandex.ru

; <<> DiG 9.18.33 <<> www.yandex.ru
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 28785
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;www.yandex.ru.                IN      A

;; ANSWER SECTION:
www.yandex.ru.                479     IN      A      5.255.255.77
www.yandex.ru.                479     IN      A      77.88.55.88
www.yandex.ru.                479     IN      A      77.88.44.55

;; Query time: 93 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Sat Sep 12 10:43:50 UTC 2026
;; MSG SIZE rcvd: 90
```

Рис. 4: Запрос dig www.yandex.ru

Конфигурирование кэширующего DNS-сервера

Анализируем содержимое /etc/resolv.conf и /etc/named.conf.

```
===== /etc/resolv.conf =====
# Generated by NetworkManager
search mavinogradova.net
nameserver 8.8.8.8
nameserver fd17:625c:f037:2::3

===== /etc/named.conf =====
//
// named.conf
//
// Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
// server as a caching only nameserver (as a localhost DNS resolver only).
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//
options {
    listen-on port 53 { 127.0.0.1; };
    listen-on-v6 port 53 { ::1; };
    directory      "/var/named";
    dump-file       "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secroots-file   "/var/named/data/named.secroots";
    recursing-file  "/var/named/data/named.recursing";
    allow-query     { localhost; };
}
```

Рис. 5: Содержимое resolv.conf и named.conf

Анализируем содержимое /var/named/named.ca.

```
==== /var/named/named.ca ====
;
; This file holds the information on root name servers needed to
; initialize cache of Internet domain name servers
; (e.g. reference this file in the "cache . <file>"
; configuration file of BIND domain name servers).
;
; This file is made available by InterNIC
; under anonymous FTP as
;   file           /domain/named.cache
;   on server      FTP.INTERNIC.NET
; -OR-             RS.INTERNIC.NET
;
; last update:      December 20, 2023
; related version of root zone:  2023122001
;
; FORMERLY NS.INTERNIC.NET
;
;
; 3600000      NS      A.ROOT-SERVERS.NET.
A.ROOT-SERVERS.NET. 3600000      A      198.41.0.4
A.ROOT-SERVERS.NET. 3600000      AAAA  2001:503:ba3e::2:30
;
; FORMERLY NS1.ISI.EDU
;
;
; 3600000      NS      B.ROOT-SERVERS.NET.
B.ROOT-SERVERS.NET. 3600000      A      170.247.170.2
B.ROOT-SERVERS.NET. 3600000      AAAA  2001:1b8:10::b
;
; FORMERLY C.PSI.NET
;
;
; 3600000      NS      C.ROOT-SERVERS.NET.
C.ROOT-SERVERS.NET. 3600000      A      192.33.4.12
C.ROOT-SERVERS.NET. 3600000      AAAA  2001:500:2::c
;
; FORMERLY TERP.UMD.EDU
;
;
; 3600000      NS      D.ROOT-SERVERS.NET.
D.ROOT-SERVERS.NET. 3600000      A      199.7.91.13
D.ROOT-SERVERS.NET. 3600000      AAAA  2001:500:2d::d
;
; FORMERLY NS.NASA.GOV
;
;
; 3600000      NS      E.ROOT-SERVERS.NET.
E.ROOT-SERVERS.NET. 3600000      A      192.203.230.10
```

Рис. 6: Содержимое named.ca

Анализируем содержимое named.localhost и named.loopback.

```
===== /var/named/named.localhost =====
$TTL 1D
@      IN SOA  @ rname.invalid. (
                                0      ; serial
                                1D      ; refresh
                                1H      ; retry
                                1W      ; expire
                                3H )    ; minimum

      NS      @
      A      127.0.0.1
      AAAA    ::1

===== /var/named/named.loopback =====
$TTL 1D
@      IN SOA  @ rname.invalid. (
                                0      ; serial
                                1D      ; refresh
                                1H      ; retry
                                1W      ; expire
                                3H )    ; minimum

      NS      @
      A      127.0.0.1
      AAAA    ::1
      PTR     localhost.
```

Рис. 7: Содержимое named.localhost и named.loopback

Запускаем DNS-сервер и включаем его в автозапуск.

```
[root@server.navinogradova.net ~]# systemctl start named  
[root@server.navinogradova.net ~]# systemctl enable named  
Created symlink '/etc/systemd/system/multi-user.target.wants/named.service' → '/usr/lib/systemd/system/named.service'.
```

Рис. 8: Запуск и включение автозапуска named

Повторно выполняем запрос dig www.yandex.ru.

```
[root@server.mavinogradova.net ~]# dig www.yandex.ru

; <<>> DiG 9.18.33 <<>> www.yandex.ru
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 35997
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags;; udp: 1232
;; QUESTION SECTION:
;www.yandex.ru.                IN      A

;; ANSWER SECTION:
www.yandex.ru.                513     IN      A      77.88.44.55
www.yandex.ru.                513     IN      A      5.255.255.77
www.yandex.ru.                513     IN      A      77.88.55.88

;; Query time: 91 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Sat Sep 12 10:50:18 UTC 2026
;; MSG SIZE rcvd: 90
```

Рис. 9: Повторный запрос dig www.yandex.ru

Запрос к локальному DNS-серверу — dig @127.0.0.1 www.yandex.ru.

```
[root@server.mavinogradova.net ~]# dig @127.0.0.1 www.yandex.ru

; <<>> DiG 9.18.33 <<>> @127.0.0.1 www.yandex.ru
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 58146
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 4c57ff2c5a65bdbf010000006aa52e764578e26ae6246d57 (good)
;; QUESTION SECTION:
;www.yandex.ru.                IN      A

;; Query time: 4126 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Sat Sep 12 10:50:30 UTC 2026
;; MSG SIZE rcvd: 70
```

Рис. 10: Запрос dig 127.0.0.1 www.yandex.ru

Делаем DNS-сервер сервером по умолчанию: настраиваем соединение eth0.

```
[root@server.mavinogradova.net ~]# nmcli connection edit eth0
===| nmcli interactive connection editor |===
Editing existing '802-3-ethernet' connection: 'eth0'
Type 'help' or '?' for available commands.
Type 'print' to show all the connection properties.
Type 'describe [<setting>.<prop>]' for detailed property description.
You may edit the following settings: connection, 802-3-ethernet (ether
e, link, tc, proxy
```

Рис. 11: Редактирование соединения eth0

Соединение System eth0 в Rocky Linux 10 отсутствует.

```
nmcli> remove ipv4.dns
nmcli> set ipv4.ignore-auto-dns yes
nmcli> set ipv4.dns 127.0.0.1
nmcli> save
Connection 'eth0' (31ef6182-2c55-44b3-8944-a2af7d10d59a) successfully updated.
nmcli> quit
```

Рис. 12: Соединение System eth0 отсутствует

Перезапускаем NetworkManager и проверяем `/etc/resolv.conf`.

```
[root@server.mavinogradova.net ~]# nmcli connection edit System\ eth0  
Error: Unknown connection 'System eth0'.
```

Рис. 13: Перезапуск NetworkManager и проверка `resolv.conf`

Вносим изменения в /etc/named.conf: listen-on и allow-query.

```
[root@server.mavinogradova.net ~]# systemctl restart NetworkManager
[root@server.mavinogradova.net ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search mavinogradova.net
nameserver 127.0.0.1
nameserver fd17:625c:f037:2::3
```

Рис. 14: Изменение параметров listen-on и allow-query в named.conf

Разрешаем работу с DNS в межсетевом экране.

```
options {  
    listen-on port 53 { 127.0.0.1; any; };  
    listen-on-v6 port 53 { ::1; };  
    directory      "/var/named";  
    dump-file      "/var/named/data/cache_dump.db";  
    statistics-file "/var/named/data/named_stats.txt";  
    memstatistics-file "/var/named/data/named_mem_stats.txt";  
    secroots-file  "/var/named/data/named.secroots";  
    recursing-file  "/var/named/data/named.recursing";  
    allow-query     { localhost; 192.168.0.0/16; };  
}
```

Рис. 15: Разрешение DNS в firewall

Проверяем прослушивание порта 53 при помощи lsof.

```
[root@server.mavinogradova.net ~]# firewall-cmd --add-service=dns
success
[root@server.mavinogradova.net ~]# firewall-cmd--add-service=dns--permanent
bash: firewall-cmd--add-service=dns--permanent: command not found...
^C
[root@server.mavinogradova.net ~]# firewall-cmd --add-service=dns --permanent
success
```

Рис. 16: Проверка прослушивания порта 53 через lsof

Конфигурирование первичного DNS-сервера

Копируем шаблон зон named.rfc1912.zones и переименовываем в mavinogradova.net.

```
[root@server.mavinogradova.net ~]# lsuf | grep UDP
lsuf: WARNING: can't stat() /usr/porta1/file system /run/user/42/doc
Output information may be incomplete.
avahi-dae 1865          avahi 12u  IPv4  9634      0t0  UDP *.mdns
avahi-dae 1865          avahi 13u  IPv6  9635      0t0  UDP *.mdns
chronyd 1866           chrony 4u   IPv4  9864      0t0  UDP localhost:323
chronyd 1866           chrony 5u   IPv6  9865      0t0  UDP localhost:323
chronyd 1866           chrony 8u   sock  8,9      0t0  28366 protocol: UDP
NetworkMa 9626          root 33u  IPv4  45412     0t0  UDP server.mavinogradova.net:bootpc->.gat
NetworkMa 9626 9638 psol-span root 33u  IPv4  45412     0t0  UDP server.mavinogradova.net:bootpc->.gat
NetworkMa 9626 9639 gbus   root 33u  IPv4  45412     0t0  UDP server.mavinogradova.net:bootpc->.gat
named 10136            named 25u  IPv4  56921     0t0  UDP localhost:domain
named 10136            named 26u  IPv4  56922     0t0  UDP localhost:domain
named 10136            named 31u  IPv4  56925     0t0  UDP server.mavinogradova.net:domain
named 10136            named 32u  IPv4  56926     0t0  UDP server.mavinogradova.net:domain
named 10136            named 35u  IPv4  51799     0t0  UDP server.mavinogradova.net:domain
named 10136            named 36u  IPv4  51800     0t0  UDP server.mavinogradova.net:domain
named 10136            named 39u  IPv6  51803     0t0  UDP localhost:domain
named 10136            named 48u  IPv6  51804     0t0  UDP localhost:domain
named 10136 10137 isc-net-0 named 25u  IPv4  56921     0t0  UDP localhost:domain
named 10136 10137 isc-net-0 named 26u  IPv4  56922     0t0  UDP localhost:domain
named 10136 10137 isc-net-0 named 31u  IPv4  56925     0t0  UDP server.mavinogradova.net:domain
named 10136 10137 isc-net-0 named 32u  IPv4  56926     0t0  UDP server.mavinogradova.net:domain
named 10136 10137 isc-net-0 named 35u  IPv4  51799     0t0  UDP server.mavinogradova.net:domain
named 10136 10137 isc-net-0 named 36u  IPv4  51800     0t0  UDP server.mavinogradova.net:domain
named 10136 10137 isc-net-0 named 39u  IPv6  51803     0t0  UDP localhost:domain
named 10136 10137 isc-net-0 named 48u  IPv6  51804     0t0  UDP localhost:domain
named 10136 10138 isc-net-0 named 25u  IPv4  56921     0t0  UDP localhost:domain
named 10136 10138 isc-net-0 named 26u  IPv4  56922     0t0  UDP localhost:domain
named 10136 10138 isc-net-0 named 31u  IPv4  56925     0t0  UDP server.mavinogradova.net:domain
named 10136 10138 isc-net-0 named 32u  IPv4  56926     0t0  UDP server.mavinogradova.net:domain
named 10136 10138 isc-net-0 named 35u  IPv4  51799     0t0  UDP server.mavinogradova.net:domain
named 10136 10138 isc-net-0 named 36u  IPv4  51800     0t0  UDP server.mavinogradova.net:domain
```

Рис. 17: Копирование шаблона зон

Подключаем файл зон в /etc/named.conf.

```
[root@server.mavinogradova.net ~]# cp /etc/named.rfc1912.zones /etc/named/  
[root@server.mavinogradova.net ~]# cd /etc/named  
[root@server.mavinogradova.net named]# mv /etc/named/named.rfc1912.zones /etc/named/mavinogradova.net
```

Рис. 18: Подключение файла зон в named.conf

Прописываем прямую и обратную зоны, остальное удаляем.

```
include "/etc/named/mavinogradova.net";  
|
```

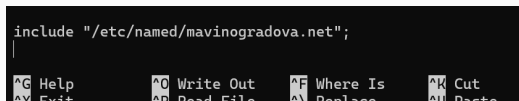


Рис. 19: Файл /etc/named/mavinogradova.net после правки

Создаём каталоги master/fz и master/rz.

```
// disable-empty-zone "."; into options
//

zone "mavinogradova.net" IN {
    type primary;
    file "master/fz/mavinogradova.net";
    allow-update { none; };
};

zone "1.168.192.in-addr.arpa" IN {
    type primary;
    file "master/rz/192.168.1";
    allow-update { none; };
};
```

Рис. 20: Создание каталогов master/fz и master/rz

Копируем шаблон прямой зоны named.localhost.

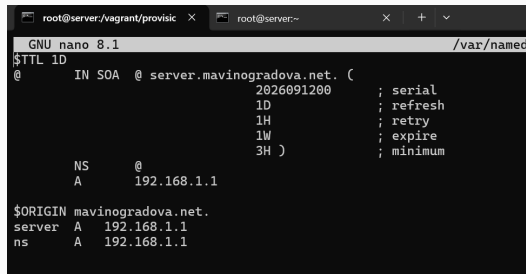
```
[root@server.mavinogradova.net named]# mkdir -p /var/named/master/fz
[root@server.mavinogradova.net named]# mkdir -p /var/named/master/rz
[root@server.mavinogradova.net named]# ls -la /var/named/master/
total 0
drwxr-xr-x. 4 root root   26 Sep 12 11:33 .
drwxrwx--T. 6 root named 141 Sep 12 11:33 ..
drwxr-xr-x. 2 root root    6 Sep 12 11:33 fz
drwxr-xr-x. 2 root root    6 Sep 12 11:33 rz
```

Рис. 21: Копирование шаблона прямой зоны

```
[root@server.mavinogradova.net named]# cp /var/named/named.localhost /var/named/master/fz/
[root@server.mavinogradova.net named]# cd /var/named/master/fz/
[root@server.mavinogradova.net fz]# mv named.localhost mavinogradova.net
[root@server.mavinogradova.net fz]# ls -la /var/named/master/fz/
total 4
drwxr-xr-x. 2 root root 31 Sep 12 11:35 .
drwxr-xr-x. 4 root root 26 Sep 12 11:33 ..
-rw-r-----. 1 root root 152 Sep 12 11:34 mavinogradova.net
```

Рис. 22: Правка файла прямой зоны

Копируем шаблон обратной зоны named.loopback.



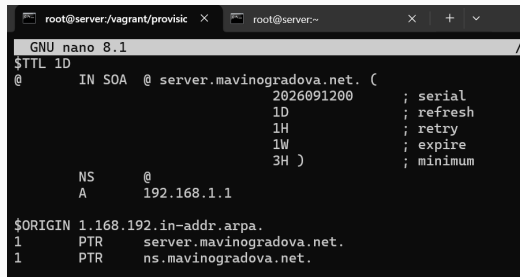
```
GNU nano 8.1 /var/named
$TTL 1D
@      IN SOA  @ server.mavinogradova.net. (
                                2026091200    ; serial
                                1D              ; refresh
                                1H              ; retry
                                1W              ; expire
                                3H )           ; minimum

      NS      @
      A       192.168.1.1

$ORIGIN mavinogradova.net.
server A 192.168.1.1
ns      A 192.168.1.1
```

Рис. 23: Копирование шаблона обратной зоны

Правка файла обратной зоны.



```
GNU nano 8.1
$TTL 1D
@      IN SOA  @ server.mavinogradova.net. (
                                2026091200      ; serial
                                1D                ; refresh
                                1H                ; retry
                                1W                ; expire
                                3H )              ; minimum

      NS      @
      A      192.168.1.1

$ORIGIN 1.168.192.in-addr.arpa.
1      PTR    server.mavinogradova.net.
1      PTR    ns.mavinogradova.net.
```

Рис. 24: Правка файла обратной зоны

Исправляем права доступа к файлам в /etc/named и /var/named.

```
[root@server.mavinogradova.net fz]# cp /var/named/named.loopback /var/named/master/rz/
[root@server.mavinogradova.net fz]# cd /var/named/master/rz/
[root@server.mavinogradova.net rz]# mv named.loopback 192.168.1
[root@server.mavinogradova.net rz]# ls -la /var/named/master/rz/
total 4
drwxr-xr-x. 2 root root 23 Sep 12 11:43 .
drwxr-xr-x. 4 root root 26 Sep 12 11:33 ..

[root@server.mavinogradova.net rz]# named-checkzone 1.168.192.in-addr.arpa /var/named/master/rz/
zone 1.168.192.in-addr.arpa/IN: loaded serial 2026091200
OK
[root@server.mavinogradova.net rz]# ls -la /var/named/master/rz/
total 4
drwxr-xr-x. 2 root root 23 Sep 12 11:43 .
drwxr-xr-x. 4 root root 26 Sep 12 11:33 ..
-rw-r-----. 1 root root 499 Sep 12 11:45 192.168.1
```

Рис. 25: Исправление прав доступа

Восстанавливаем метки SELinux и настраиваем переключатели.

```
[root@server.mavinogradova.net rz]# chown -R named:named /etc/named
[root@server.mavinogradova.net rz]# chown -R named:named /etc/named
[root@server.mavinogradova.net rz]# ls -la /etc/named/mavinogradova.net
-rw-r-----. 1 named named 731 Sep 12 11:32 /etc/named/mavinogradova.net
[root@server.mavinogradova.net rz]# ls -la /var/named/master/fz/mavinogradova.net
-rw-r-----. 1 root root 462 Sep 12 11:41 /var/named/master/fz/mavinogradova.net
[root@server.mavinogradova.net rz]# ls -la /var/named/master/rz/192.168.1
-rw-r-----. 1 root root 499 Sep 12 11:45 /var/named/master/rz/192.168.1
[root@server.mavinogradova.net rz]# killall chown
chown: no process found
[root@server.mavinogradova.net rz]# chown -R named:named /var/named
[root@server.mavinogradova.net rz]# ls -la /var/named/master/fz/mavinogradova.net
-rw-r-----. 1 named named 462 Sep 12 11:41 /var/named/master/fz/mavinogradova.net
[root@server.mavinogradova.net rz]# ls -la /var/named/master/rz/192.168.1
-rw-r-----. 1 named named 499 Sep 12 11:45 /var/named/master/rz/192.168.1
```

Рис. 26: Восстановление меток SELinux и настройка переключателей

Перезапускаем named и смотрим логи во втором терминале.

```
[root@server.mavingradova.set rz]# restorecon -v /etc
Relabeled /etc/lw/devices/system.devices from system_u:object_r:lw_metadata_t:s0 to system_u:object_r:lw_etc_t:s0
Relabeled /etc/lw/devices/bioskey/system.devices-2020011_165210.5000 from system_u:object_r:lw_metadata_t:s0 to system_u:object_r:lw_etc_t:s0
Relabeled /etc/NetworkManager/system-connections/eth1.nmconnection from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:NetworkManager.st
c_rw_t:s0
Relabeled /etc/foomatic/hashes.d/hashes.new from system_u:object_r:etc_runtime_t:s0 to system_u:object_r:etc_t:s0
[root@server.mavingradova.set rz]# restorecon -v /var/named
bash: restorecon-v: command not found...
[root@server.mavingradova.set rz]# restorecon -v /var/named
[root@server.mavingradova.set rz]# getsebool -a | grep named
named_tcp_bind_http_port == off
named_write_master_zones == on
[root@server.mavingradova.set rz]# setsebool named_write_master_zones 1
[root@server.mavingradova.set rz]# setsebool -P named_write_master_zones 1
bash: setsebool-P: command not found...
[root@server.mavingradova.set rz]# setsebool -P named_write_master_zones 1
[root@server.mavingradova.set rz]# systemctl restart named
```

Рис. 27: Перезапуск named и логи во втором терминале

Анализ работы DNS-сервера

Получаем описание DNS-зоны при помощи dig.

```
Sep 12 18:03:24 server.mavingradova.net named[10975]: configuring command channel from '/etc/rndc.hkey'
Sep 12 18:03:24 server.mavingradova.net named[10975]: command channel listening on *10953
Sep 12 18:03:24 server.mavingradova.net named[10975]: managed-keys-zone: loaded serial 0
Sep 12 18:03:24 server.mavingradova.net named[10975]: zone 0.0.0.0-addr.arpa/IN: loaded serial 0
Sep 12 18:03:24 server.mavingradova.net named[10975]: zone 0.0.0.0-127.0.0.1.arpa/IN: loaded serial 0
Sep 12 18:03:24 server.mavingradova.net named[10975]: zone localhost/IN: loaded serial 0
Sep 12 18:03:24 server.mavingradova.net named[10975]: zone localhost.localdomain/IN: loaded serial 0
Sep 12 18:03:24 server.mavingradova.net named[10975]: zone 1.168.192.1-in-addr.arpa/IN: Loaded serial 2026091200
Sep 12 18:03:24 server.mavingradova.net named[10975]: zone 1.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.f.ip6.arpa/IN: loaded serial 0
Sep 12 18:03:24 server.mavingradova.net named[10975]: zone mavingradova.net/IN: loaded serial 2026091200
Sep 12 18:03:24 server.mavingradova.net named[10975]: all zones loaded
Sep 12 18:03:24 server.mavingradova.net named[10975]: running
Sep 12 18:03:24 server.mavingradova.net systemd[1]: Started named.service - Berkeley Internet Name Domain (DNS).
Subject: A fatal error has occurred: named service has finished successfully
Defined by: systemd
Support: https://redhat-magazine.com/post/yum-support
A start job for unit named.service has finished successfully.
```

```
The Job Identifier is 36062
Sep 12 18:03:34 server.mavingradova.net named[10975]: managed-keys-zone: Unable to fetch DNSKEY set ": timed out
Sep 12 18:03:34 server.mavingradova.net named[10975]: recursive parsing query complete: time out
Sep 12 18:07:05 server.mavingradova.net named[10975]: Client 800.73.801.62#4048 [127.0.0.1] 193997 (mavingradova.net): transfer of 'mavingradova.net/IN':
DSX started (serial 2026091200)
Sep 12 18:07:05 server.mavingradova.net named[10975]: client 800.73.801.62#4048 [127.0.0.1] 193997 (mavingradova.net): transfer of 'mavingradova.net/IN':
AXFR received: 1 messages, 0 records, 379 bytes, 0.009 secs (13888 bytes/sec) (serial 2026091200)
Sep 12 18:07:05 server.mavingradova.net named[10975]: managed-keys-zone: unable to fetch DNSKEY set ": timed out
Sep 12 18:07:30 server.mavingradova.net named[10975]: managed-keys-zone: Initializing automatic trust anchor management for zone ": DNSKEY ID 20936
is now trusted, waiting the normal 30-day waiting period.
Sep 12 18:07:30 server.mavingradova.net named[10975]: managed-keys-zone: Initializing automatic trust anchor management for zone ": DNSKEY ID 36069
is now trusted, waiting the normal 30-day waiting period.
```

Рис. 28: Получение описания DNS-зоны

Анализируем работу DNS-сервера при помощи host.

```
[root@server.mavogradova.net rz]# dig ns.mavogradova.net

;<<< DiG 9.18.33 <<< ns.mavogradova.net
;; global options: +cmd
;; Got answer:
-->>>HEADER<<< opcode: QUERY, status: NOERROR, id: 2499
;; Flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: afb6f5cb2b0d62a010080806aa54024ab3e97c549faaff2 (good)
;; QUESTION SECTION:
;ns.mavogradova.net.                IN      A

;; ANSWER SECTION:
ns.mavogradova.net.  86400   IN      A      192.168.1.1

;; Query time: 4 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Sat Sep 12 12:09:56 UTC 2026
;; MSG SIZE rcvd: 93

[root@server.mavogradova.net rz]# host -l mavogradova.net
mavogradova.net name server mavogradova.net.
mavogradova.net has address 192.168.1.1
ns.mavogradova.net has address 192.168.1.1
server.mavogradova.net has address 192.168.1.1
[root@server.mavogradova.net rz]# host -a mavogradova.net
Trying "mavogradova.net"
;<<<HEADER<<< opcode: QUERY, status: NOERROR, id: 68
;; Flags: qr aa rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;mavogradova.net.                IN      ANY

;; ANSWER SECTION:
mavogradova.net.  86400   IN      SOA     mavogradova.net. server.mavogradova.net. 2026091200 86400 3600 604800 10000
mavogradova.net.  86400   IN      NS      mavogradova.net.
mavogradova.net.  86400   IN      A      192.168.1.1

Received 188 bytes from 127.0.0.1#53 in 12 ms
[root@server.mavogradova.net rz]# host -t A mavogradova.net
mavogradova.net has address 192.168.1.1
[root@server.mavogradova.net rz]# host -t PTR 192.168.1.1
1.1.168.192.in-addr.arpa domain name pointer server.mavogradova.net.
1.1.168.192.in-addr.arpa domain name pointer ns.mavogradova.net.
```

Рис. 29: Анализ работы DNS-сервера через host

Внесение изменений в настройки внутреннего окружения

Копируем конфигурационные файлы DNS в каталог provision.

```
[root@server.navinogradova.net r2]# cd /vagrant
[root@server.navinogradova.net vagrant]# mkdir -p /vagrant/provision/server/dns/etc/named
bash: mkdir -p: command not found...
[root@server.navinogradova.net vagrant]# mkdir -p /vagrant/provision/server/dns/etc/named
[root@server.navinogradova.net vagrant]# mkdir -p /vagrant/provision/server/dns/var/named/master/
[root@server.navinogradova.net vagrant]# cp -R /etc/named.conf /vagrant/provision/server/dns/etc/
bash: cp -R: command not found...
[root@server.navinogradova.net vagrant]# cp -R /etc/named.conf /vagrant/provision/server/dns/etc/
[root@server.navinogradova.net vagrant]# cp -R /etc/named/* /vagrant/provision/server/dns/etc/named/
[root@server.navinogradova.net vagrant]# cp -R /var/named/master/* /vagrant/provision/server/dns/var/named/master/
[root@server.navinogradova.net vagrant]# ls -R /vagrant/provision/server/dns/
/vagrant/provision/server/dns/:
named.conf

/vagrant/provision/server/dns/etc:
named.conf

/vagrant/provision/server/dns/etc/named:
navinogradova.net

/vagrant/provision/server/dns/var:

/vagrant/provision/server/dns/var/named:

/vagrant/provision/server/dns/var/named/master:
[root@server.navinogradova.net vagrant]# cp -R /var/named/master/. /vagrant/provision/server/dns/var/named/master/
[root@server.navinogradova.net vagrant]# ls -R /vagrant/provision/server/dns/
/vagrant/provision/server/dns/:
named.conf

/vagrant/provision/server/dns/etc:
named.conf

/vagrant/provision/server/dns/etc/named:
navinogradova.net

/vagrant/provision/server/dns/var:

/vagrant/provision/server/dns/var/named:
master/
```

Рис. 30: Копирование конфигурационных файлов DNS в каталог provision

Создаём скрипт dns.sh для Vagrant.

```
root@server:~/vagrant/provisic x root@server:~ x + v
GNU nano 8.1
#!/bin/bash

echo "Provisioning script $0"

echo "Install needed packages"
dnf -y install bind bind-utils

echo "Copy configuration files"
cp -R /vagrant/provision/server/dns/etc/* /etc
cp -R /vagrant/provision/server/dns/var/named/* /var/named

chown -R named:named /etc/named
chown -R named:named /var/named

restorecon -vR /etc
restorecon -vR /var/named

echo "Configure firewall"
firewall-cmd --add-service=dns
firewall-cmd --add-service=dns --permanent

echo "Tuning SELinux"
setsebool named_write_master_zones 1
setsebool -P named_write_master_zones 1

echo "Change dns server address"
nmcli connection edit eth0 <<EOF
remove ipv4.dns
set ipv4.ignore-auto-dns yes
set ipv4.dns 127.0.0.1
save
quit
EOF
systemctl restart NetworkManager

echo "Start named service"
systemctl enable named
systemctl start named
```

Добавляем строку `server.vm.provision "server dns"` в Vagrantfile.

```
## Server configuration
config.vm.define "server", autostart: false do |server|
  server.vm.box = "rockylinux10"
  server.vm.hostname = 'server'

  server.vm.boot_timeout = 1440

  server.ssh.insert_key = false
  server.ssh.username = 'vagrant'
  server.ssh.password = 'vagrant'

  server.vm.network :private_network,
    ip: "192.168.1.1",
    virtualbox____intnet: true

  server.vm.provider :virtualbox do |virtualbox|
    virtualbox.customize ["modifyvm", :id, "--vrde", "on"]
    virtualbox.customize ["modifyvm", :id, "--vrdeport", "3391"]
  end

  server.vm.provision "server dummy",
    type: "shell",
    preserve_order: true,
    path: "provision/server/01-dummy.sh"

  server.vm.provision "server dns",
    type: "shell",
    preserve_order: true,
    path: "provision/server/dns.sh"
end
```

Рис. 32: Фрагмент Vagrantfile с добавленной строкой

Выводы

Приобретены практические навыки по установке и конфигурированию DNS-сервера, усвоены принципы работы системы доменных имён.

На виртуальной машине `server`:

- установлены пакеты `bind` и `bind-utils`;
- сконфигурирован кэширующий DNS-сервер;
- сконфигурирован первичный DNS-сервер с прямой зоной `mavinogradova.net` и обратной зоной `1.168.192.in-addr.arpa`;
- работа DNS-сервера проверена утилитами `dig` и `host`;
- написан скрипт `dns.sh` для Vagrant, фиксирующий действия по установке и конфигурированию DNS-сервера, и внесены соответствующие изменения в `Vagrantfile`.